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ORIGINAL RESEARCH

Highly efficient dissolution of xylan in ionic liquid-based deep eutectic solvents

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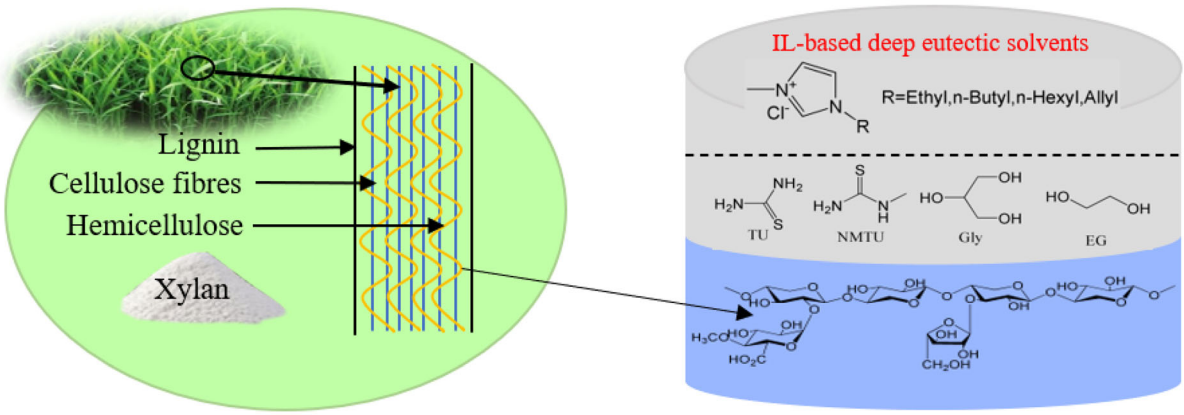
Abstract Development of efficient solvents to dissolve lignocellulose and its main components is of significant importance for the utilization/transformation of lignocellulose. Herein, several ionic liquid (IL)-based deep eutectic solvents (DESs) were prepared using four ILs as the hydrogen bond acceptors and four different hydrogen bond donors, including thiourea (TU), N-methylthiourea (NMTU), glycerol (Gly), and ethylene glycol (EG). Using the dissolution of xylan (a typical type of hemicellulose) as the example, it was observed that the synthesized IL-based DESs could efficiently dissolve xylan. Especially in the DES formed from 1-allyl-3-methylimidazolium chloride and EG, the xylan solubility could reach as high as 40.4 wt% at 343.15 K. More importantly, the solubility of xylan in IL-based DESs was much higher than that in the corresponding IL. The enhancement effect might be ascribed to that the formation of DESs could significantly decrease the viscosity and increase the hydrogen bond accepting ability. Additionally, the regeneration of the xylan from the DESs could be easily realized using ethanol as the anti-solvent, and the regenerated xylan kept a high consistency with the original xylan.

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Graphic abstract Ionic liquid-based deep eutectic solvents were designed for highly efficient dissolution of xylan owing to the decreased viscosity and the increased hydrogen-bond accepting ability.



Keywords Lignocellulose · Dissolution of xylan · Deep eutectic solvents · Ionic liquids · Kamlet-taft empirical parameters

Introduction

With the gradual diminishing of nonrenewable fossil resources (i.e., coal, petroleum, and natural gas), interest has been devoted on the pursuit of alternative resources. In this context, lignocellulose, a renewable and abundant resource, has gained significant attention as the alternative of fossil resources to produce various valuable chemicals and functional materials (Mohanty et al. 2018; Schutyser et al. 2018; Sudarsanam et al. 2019; Sun et al. 2018; Zhang et al. 2017). Generally, to efficiently utilize or convert lignocellulose, its dissolution in some suitable solvent is an important issue.

Traditional organic solvents generally have high volatility to cause environmental problems, and thus they are not good candidate for the dissolution of lignocellulose and its main components (i.e., cellulose, lignin, and hemicellulose) although some of them could be utilized to dissolve lignocellulose and its components with the satisfactory solubility (Hao et al. 2015; Huang et al. 2016; Xue et al. 2016). To replace the volatile organic solvents, ionic liquids (ILs) have been utilized as functional green solvents for the dissolution of lignocellulose and its components owing to the unique advantages of ILs (e.g., extremely low vapor pressure, large liquid range, tunable properties) (Cui et al. 2016; Hallett and Welton 2011). Good solubility of lignocellulose and its components can be achieved using ILs as the solvents (King et al. 2011; Lethesh et al. 2020; Rashid et al. 2016; Swatloski et al. 2002; Xu et al. 2017). For example, Yuan et al. (2019) found that the highest solubility of bamboo hemicellulose in 1-butyl-3-methylimidazolium halide based-ILs was 18.8 g/100 g (15.8 wt%, in [BMIM]Cl) at 120 °C. Liu et al. (2012) synthesized a series of room temperature ILs, in which cholinium acts as the cation and amino acids as the anions, and the solubility of xylan in these ILs ranged from < 1 to 85 mg g^{-1} (< 0.1–7.8 wt%). In addition, Froschauer et al. (2012) described a possibility of selective dissolution of xylan using dialkyl phosphate-related ionic liquids. However, the high viscosity of common used ILs often decreases their capability to dissolve lignocellulose and also limits their large-scale applications.

In recent years, deep eutectic solvents (DESs) have drawn numerous attention (Florindo et al. 2019; Ma et al. 2019; Ruiz-Olles et al. 2019; Smith et al. 2014; Zhang et al. 2012). DESs, composed by two components: hydrogen-bond donor (HBD) and hydrogen-bond acceptor (HBA), have similar properties like ILs, including low volatility, wide liquid range, nonflammability, and designability, etc. Therefore, DESs have been considered as the alternative of ILs in

diverse applications. Attention on the utilization of DESs to dissolve lignocellulose has been awaken in the past few years. In spite of great progress achieved for dissolution of lignocellulose in DESs, using DESs to dissolve lignocellulose is in its early stage, and the successful cases are still limited (Chen et al. 2018; Francisco et al. 2012; Liu et al. 2019; Lynam et al. 2017; Morais et al. 2018; Soares et al. 2017). Especially on the dissolution of hemicellulose, Morais et al. reported the dissolution of xylan in DES aqueous solutions, and the results showed that the highest xylan solubility (328 ± 29 mg mL^{-1}) was obtained with 66.7 wt% DES aqueous solutions at 80 °C (Morais et al. 2018); Lynam et al. (2017) obtained less than 5 wt% xylan solubility with lactic acid:choline chloride 10:1 mol ratio (LA:CC10:1). Therefore, it is highly desired to develop novel and robust DESs for the dissolution of lignocellulose and its components.

Currently, most of DESs are synthesized using quaternary ammonium salts as the HBAs. Considering that some types of ILs are quaternary ammonium salts, these ILs can be used as special HBAs to prepare DESs. Due to the strong hydrogen-bonding interaction between HBD and ILs (as the HBAs), the melting point and the viscosity of the formed DESs would be potentially decreased compared with the corresponding components (HBDs and ILs). Thus, we deduce that these IL-based DESs by combining the advantages of both ILs and DESs could probably be employed as efficient solvents to dissolve lignocellulose and its components.

On the basis of the above discussion, herein, several IL-based DESs (Fig. 1a) were designed by employing thiourea (TU), N-methylthiourea (NMTU), glycerol (Gly) and ethylene glycol (EG) as the HBDs and four ILs [i.e., 1-butyl-3-methylimidazolium chloride ([BMIM]Cl), 1-ethyl-3-methylimidazolium chloride ([EMIM]Cl), 1-hexyl-3-methylimidazolium chloride ([HMIM]Cl), 1-allyl-3-methylimidazolium chloride ([AMIM]Cl)] as the HBAs. To our delight, the prepared DESs showed good performance on the dissolution of xylan (a typical type of hemicellulose, Fig. 1b). More importantly, the solubility of xylan in these IL-based DESs was highly enhanced in comparison with the results in the corresponding ILs.

Material and methods

Material

Xylan (from corn cob) (Fig. 1b) was purchased from Beijing Innochem Technology Co., Ltd. (Beijing, China). 1-Butyl-3-methylimidazolium chloride ([BMIM]Cl, 97%), 1-ethyl-3-methylimidazolium chloride ([EMIM]Cl, 98%), 1-hexyl-3-methylimidazolium chloride ([HMIM]Cl, 98%), and 1-allyl-3-methylimidazolium chloride ([AMIM]Cl, 96%) were obtained from Shanghai Macklin Biochemical Co., Ltd (Shanghai, China). Nile red and 4-nitroaniline were purchased from J&K Scientific. Thiourea (TU), N-methylthiourea (NMTU), ethylene glycol (EG), and glycerol (Gly) were purchased from Shanghai Aladdin Biochemical Co., Ltd (Shanghai, China).

Preparation and characterization of DESs

In a typical process, ILs as HBA, and TU, NMTU, Gly, and EG as HBD were firstly mixed at a molar ratio of 1:1, 2:1 or 3:1, respectively. These DESs were prepared by mixing the two components in a glass vial with a mechanical stirrer at 500 rpm and 363.15 K until a transparent liquid was obtained. Finally, all DESs were stored under vacuum at 333.15 K for 24 h to remove the contained water.

The melting points of DESs were measured via different scanning calorimetry (DSC) conducted on a Q2000 DSC (TA Instruments Waters LCC, USA) at a heating rate of 10 °C min^{-1} under nitrogen atmosphere, and all the samples for DSC measurements were sealed in Al pans.

The measurements of densities and viscosities for DESs were performed at different temperature, including 298.15, 308.15, 318.15, and 328.15 K, at atmospheric pressure, using an Anton Paar DMA 5000 M density meter with the accuracy of $\pm 5 \times 10^{-6}$ g cm^{-3} and an Anton Paar AMVn viscosity meter using a falling ball technique with an accuracy of $< 0.5\%$. Note that here, the obtained density and viscosity were the average value of four-time measurement, with a deviation of 0.1%. The ^{1}H NMR (nuclear magnetic resonance) spectra were taken on a Bruker Avance 400 MHz spectrometer. The decomposition temperatures of the prepared DESs were measured by thermogravimetric analysis (TGA) conducted on a TA Instruments TGA Q500.

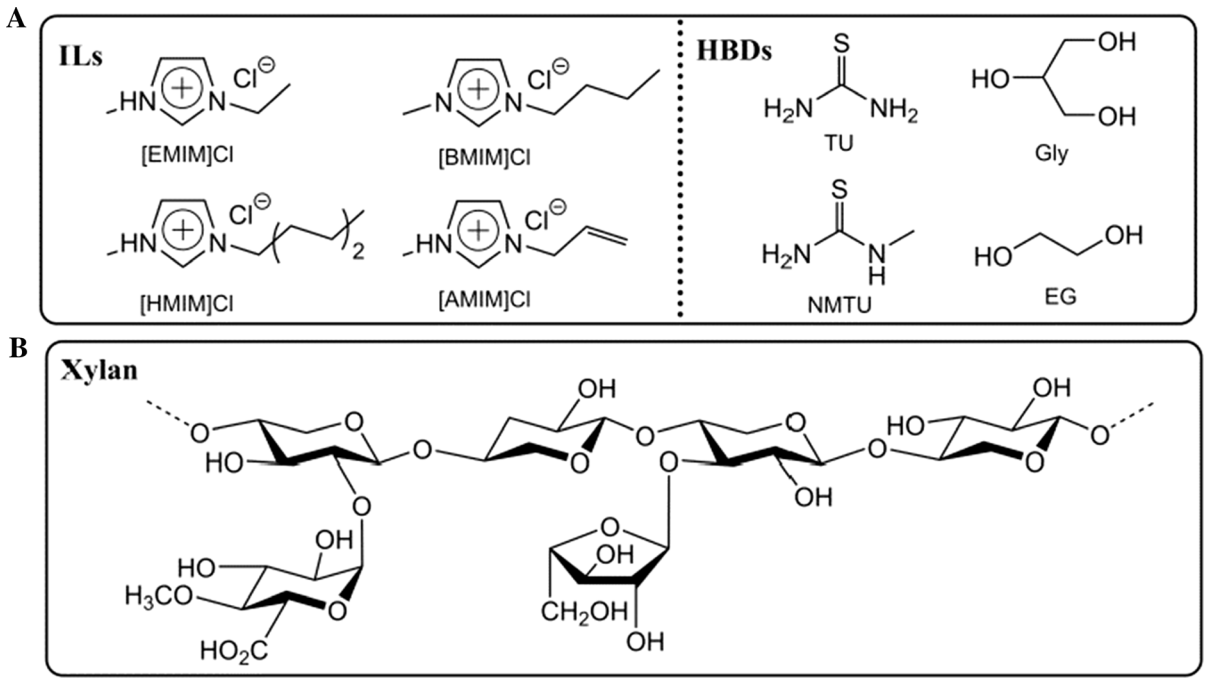


Fig. 1 The chemical structures of the synthesized DESs (**a**) and the used xylan (**b**)

Approximately 5 mg of sample was heated from 30 to 350 °C at a heating rate of 10 °C min^{-1} under nitrogen flow, and all samples were placed in a platinum pans. Fourier transform infrared (FT-IR) spectra were tested on a Prestige-21 FT-IR Spectroscopy (Shimadzu, Japan) in the range of 400–4000 cm^{-1}. The Kamlet-Taft parameters for DESs were measured, and the detailed methods were given in the Supporting Information.

Solubilities of xylan in DESs and ionic liquids

The method for measuring the solubility of xylan referred to the method for measuring the solubility of lignin by Liu et al. (2019).

In the first step, ca. 1 g of DES was added to a 20 mL glass flask and placed in an oil bath at 323.15, 333.15, and 343.15 K, respectively. After the temperature was constant, about 30 mg of xylan sample was weighed via an electronic balance (with the precision of 0.0001 g) and added into prepared DESs. After the xylan was completely dissolved, the above process was repeated until the added xylan could not be dissolved completely in 1 h. The solubility of xylan with higher potential deviations could be achieved.

In the second step, ca. 1 g of DES was added to 20 mL glass flask and placed in an oil bath at 323.15, 333.15, and 343.15 K, respectively. After the temperature was constant, about 90% of the xylan obtained in the first step was weighed and added into prepared DESs. After being completely dissolved, the xylan was continually added with an increment size of ca. 5 mg until the xylan could not be dissolved in 1 h. The amount of xylan added was recorded and the solubility was calculated.

In addition, the solubility of xylan in four ionic liquids was also measured according to the above method.

Xylan recovered from DESs

Several DESs with high solubility of xylan were selected to regenerate xylan after dissolution. Firstly, ca. 1 g of xylan was dissolved in ca. 4 g DESs at 343.15 K. After xylan was completely dissolved, xylan was precipitated by adding three volumes of ethanol (as the DESs solution). The precipitated xylan was washed with ethanol and acetone, then the recovered solid was filtered via a sand core filter device with 0.45 μm microporous filter membrane.

The recovered xylan was dried under vacuum at 313.15 K for 2 h and weighed for subsequent analysis.

FT-IR spectra were recorded with KBr pellets on a Prestige-21 FT-IR Spectroscopy (Shimadzu, Japan), and a total of 40 scans were taken for pristine xylan and recovered xylan at a resolution of 4 cm^{-1}. ^{1}H NMR and ^{13}C NMR spectra were taken on the Bruker Avance 400 MHz spectrometer.

Characterization of the acetylation degree and uronic acid content

The acetylation degrees in the original and regenerated xylan were determined on the basis of the integration of the corresponding ^{1}H NMR resonance at about 2.11 ppm (Lee et al. 2014; Morais et al. 2018). The content of uronic acids in both samples were measured according to the reported method (Bastos et al. 2015; Blumenkrantz and Asboe-Hansen 1973; Morais et al. 2018).

Kamlet-Taft LSER method

In this work, a statistical method whereby linear solvation energy relationship (LSER) was fitted by multiple linear regression to determine whether a better correlation between solubility and Kamlet-Taft parameters according to a reported method (Ab Rani et al. 2011; Gilbert et al. 2019). Here a maximum p value (0.05) was set and the multiple linear regression was expressed by the following equation:

$$\ln S = XYZ_0 + a\alpha + b\beta + c\pi^* \quad (1)$$

where a, b and c are coefficients of each parameter and XYZ_0 is intercept. First, all 3 variables {α, β, π*} were performed a multivariate linear regression to determine the coefficients, if all 3 proved statistically significant ($p < 0.05$), the correlation was accepted. If p value of one or more values were greater than 0.01, then the 3 "pairs" ({α, β}, {α, π*}, {β, π*}) were then fitted for significance, if one or more values proved statistically significant ($p < 0.05$), the correlation was accepted. If not, then all three variables ({α}, {β}, {π*}) individually were tested. Once again, if $p < 0.05$ for all members of one of these sets, then the correlation was accepted.

Results and discussion

Synthesis and characterization of the IL-based DESs

In this work, several IL-based DESs were prepared by simply heating the corresponding ILs (as the HBAs) and HBDs (Fig. 1b) with different IL/HBD molar ratios (2:1, 1:1, 3:1), and the detail procedure had been described in the Experimental section. The synthesized DESs were subsequently characterized by various techniques, including DSC (Fig. 2a), thermogravimetric analysis (TGA, Fig. 2b and S1), ^{1}H NMR (Figs. 2c and S2), and FT-IR (Figs. 2d and S3). Meanwhile, the viscosity (Tables S1–S4), density (Tables S6–S9), and solvatochromic parameters (Table S11) of these DESs were determined for their applications.

First, DSC experiments (Fig. 2a) were conducted to verify the formation of the DESs. It was found that the melting points of DESs were far below those of the corresponding pure components. For instance, the melting points of [BMIM]Cl and TU were 65 °C and 182 °C, respectively, while the melting point of [BMIM]Cl-TU (2:1) was − 62 °C, indicating that the interaction of hydrogen bonds resulted in the decrease of melting point.

Figure 2b presented TGA curves of the synthesized DESs, which denoted their thermal stability. It showed that the decomposition temperature of most of DESs was above 205 °C, which was much higher than the temperature employed to dissolve xylan, indicating that the synthesized DESs were stable in the process of xylan dissolution.

Furthermore, the intermolecular hydrogen-bond interactions between ILs and HBDs were investigated by ^{1}H NMR spectra (Fig. 2c) and FT-IR spectra (Fig. 2d). In Fig. 2c, two peaks at 3.74 ppm and 5.50 ppm in ethylene glycol represented $–CH_2–$ and –OH, respectively. When the DES was formed between ethylene glycol and [BMIM]Cl, the H signal in the -OH group showed a chemical shift from 5.50 to 5.73 ppm, indicating the existence of hydrogen-bond interaction between ethylene glycol and [BMIM]Cl. Figure 2d showed that the stretching vibration of O–H of pure ethylene corresponded to the peak at 3402 cm^{-1}, while the FT-IR spectra of DES [AMIM]Cl-EG (2:1) showed a strong broad peak at 3390 cm^{-1}. Meanwhile, the peak at 1090 cm^{-1}

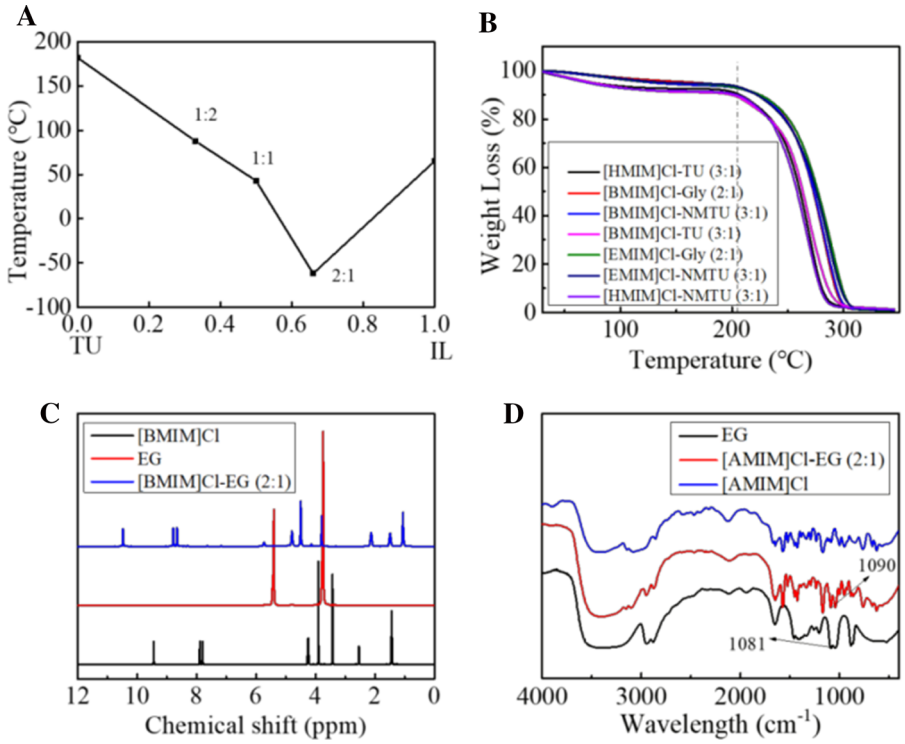


Fig. 2 The melting points of [BMIM]Cl-TU (**a**), TGA (**b**), 1H-NMR spectra (**c**) and FT-IR spectra (**d**) of the corresponding DESs

represented the stretching vibration of C–O, which was characteristic absorption of ethylene glycol. After the formation of DES, this peak shifted from 1090 to 1081 cm^{-1}, further indicating the occurrence of hydrogen-bond interaction.

The FT-IR spectra (Figure S3), ^{1}H NMR (Figure S2) and TGA curves (Figure S1) of other prepared DESs and the FT-IR spectra (Figure S4) of ILs were also carried out. Since the analysis of these results was similar to the above analysis, we would not discuss them in detail here.

Dissolution of xylan in pure ILs

The solubilities of xylan in the pure ILs (i.e., [BMIM]Cl, [EMIM]Cl, [HMIM]Cl, and [AMIM]Cl) were initially investigated (Table 1). It should be pointed out that the solubilities of xylan in [EMIM]Cl at 323.15, 333.15 and 343.15 K and in [BMIM]Cl at 323.15 and 333.15 K can not be obtained because they are solid at these temperatures (melting points of [EMIM]Cl and [BMIM]Cl are 358.15 K and 334.15 K, respectively). It was observed that the solubility of xylan in [AMIM]Cl (Table 1, entry 4) was always higher than that in [HMIM]Cl (Table 1, entry 3) and [BMIM]Cl (Table 1, entry 2) at the corresponding dissolution temperature, indicating that cations of ILs have a certain effect on the dissolution of xylan. Meanwhile, the dissolution temperature significantly affected the xylan solubility in one specific IL,

Table 1 Solubility (wt%) of xylan in ILs at different temperatures

ILs	Dissolution temperature (K)[a]		
	323.15	333.15	343.15
[EMIM]Cl	–[b]	–[b]	–[b]
[BMIM]Cl	–[b]	–[b]	1.7 (± 0.3)
[HMIM]Cl	3.1 (± 0.1)	10.5 (± 0.1)	16.8 (± 0.5)
[AMIM]Cl	13.7 (± 0.2)	20.2 (± 0.9)	23.7 (± 2.2)

Standard uncertainties u for the temperature were u(T) = 0.2 K

[a]The solubility value was the average of twice measurements, and the values in the parentheses were the half of the range of replicate results

[b]The IL was solid at this temperature, and the xylan can not be dissolved

and the solubility was improved with the increasing temperature. In the examined IL, [AMIM]Cl showed the best result owing to the smaller size of 1-allyl-3-methylimidazolium cation ($[AMIM]^+$) and its relative lower viscosity (Table S5). The smaller size was beneficial for the interaction between the IL and xylan, while the lower viscosity was helpful for mass transport (Ghaedi et al. 2017; Zhang et al. 2005). Thus, [AMIM]Cl showed the best performance for xylan dissolution, which was consistent with the dissolution of cellulose in [AMIM]-based ILs (Zhang et al. 2005).

Comparison on xylan dissolution in pure ILs and the synthesized DESs

The solubilities of xylan in the synthesized DESs and the corresponding ILs were subsequently compared (Figure S5) to confirm the advantage to form DESs. It was found that the solubilities of xylan in these prepared DESs, especially in the Gly-based DESs and EG-based DESs, were much greater than those in the corresponding individual IL. For example, the xylan solubility in [BMIM]Cl was 1.7 wt% at 70 °C, and Yuan et al. reported that the highest xylan solubility was 18.8 g/100 g (15.8 wt%) IL at 120 °C. Compared with it, the xylan solubilities in [BMIM]Cl-based DESs were greater at the same temperature. And the reason would be discussed in the following section. Meanwhile, for a specific DES, the xylan solubility increased with the increasing dissolution temperature (Figure S5). Moreover, Morais et al. reported that the highest xylan solubility (328 ± 29 mg mL^{-1}) was obtained with DES aqueous solutions (66.7 wt%) at 80 °C (Morais et al. 2018), Lynam et al. (2017) reported that lactic acid:choline chloride 10:1 mol ratio (LA:CC 10:1) showed low xylan solubility, only less than 5 wt%. Compared with those DESs, IL-based DESs showed higher solubility of xylan, especially for [AMIM]Cl-EG (2:1) with the maximum solubility of 40.4 wt% (889 ± 30 mg mL^{-1}) at 70 °C.

Effect of ILs in the synthesized DESs on xylan solubility

For the TU-based DESs with a IL/HBD molar ratio of 2:1 [i.e., [BMIM]Cl-TU (2:1), [HMIM]Cl-TU (2:1) and [AMIM]Cl-TU (2:1)], the solubility of xylan in [AMIM]Cl-TU (2:1) was higher than that in other two DESs at all the three determined temperature (Fig. 3a). It was shown that the xylan solubility was extremely low in [BMIM]Cl-based DESs (less than 3 wt% at 323.15 K and 333.15 K) owing to its too high viscosity (9666.4 MPa s). In comparison, the viscosity of [AMIM]Cl-TU (2:1) was only 110.28 MPa s, which was beneficial for xylan dissolution. Furthermore, when the IL/HBD molar ratio increased from 2:1 to 3:1, the solubilities of xylan in the three DESs ([BMIM]Cl-TU (3:1), [HMIM]Cl-TU (3:1), and [AMIM]Cl-TU (3:1)) increased to some extent (Fig. 3b) because of the decreased viscosity of DESs with the IL/HBD molar ratio of 3:1 (Table S1). It should be pointed out that [EMIM]Cl-TU-based DESs were not determined because they were generally solid at the examined temperature.

Similarly, for NMTU-based DESs with a IL/HBD molar ratio of 2:1, [AMIM]Cl-NMTU (2:1) showed the best performance on xylan dissolution (Fig. 3c) owing to its lowest viscosity (Table S2). However, when the IL/HBD molar ratio increased from 2:1 to 3:1, the solubility of xylan only increased slightly owing to the slight decrease of viscosity (Table S2).

In addition, similar phenomenon was achieved when the HBD was EG or Gly. As shown in Fig. 3e–h, the DESs formed using [AMIM]Cl had higher solubility of xylan than other DESs generated using other three ILs. Meanwhile, the solubility of xylan also slightly increased with the increasing IL/HBD molar ratio from 1:1 to 2:1 for EG or Gly-based DESs.

From Fig. 3, it could be concluded that the type of ILs could affect the solubility of xylan, and the solubility of xylan in [AMIM]Cl-based DESs was the highest among the DESs with the same HBD, which was probably caused by the lower viscosity of the [AMIM]Cl-based DESs. Meanwhile, for all the determined DESs, the xylan solubility improved with the increasing dissolution temperature from 323.15 to 343.15 K.

Effect of HBD in the synthesized DESs on xylan solubility

As shown in Fig. 3, it can be found that the HBDs in the synthesized DESs would affect the solubility of xylan. Thus, the role of HBDs on the solubilities of xylan was further discussed. It was obvious that the solubilities of xylan in EG-based DESs and Gly-based

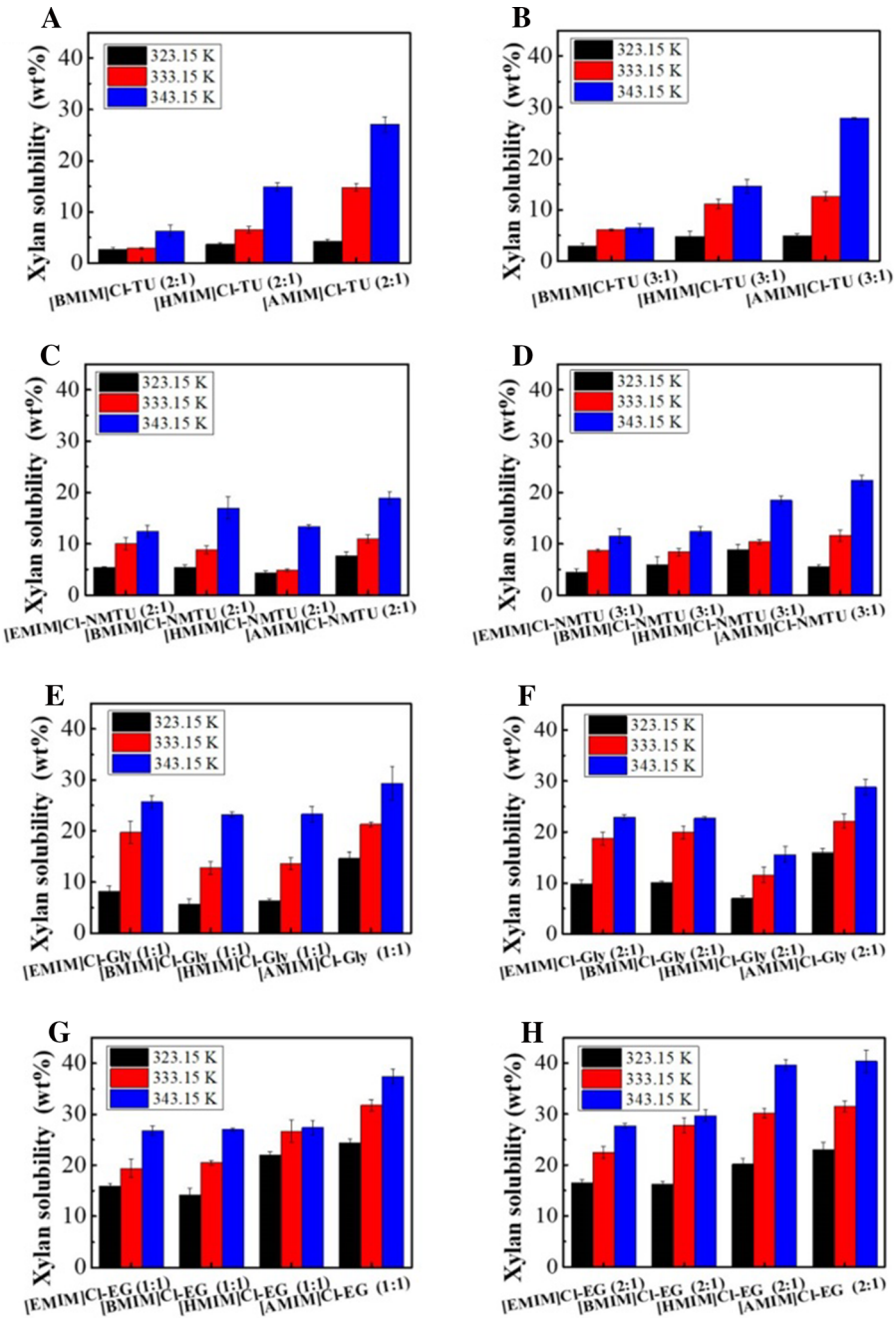


Fig. 3 The solubilities of xylan in the prepared DESs

DESs are much greater than that in TU-based DESs and NMTU-based DESs (Fig. 4). For instance, the solubility of xylan in [BMIM]Cl-EG (2:1) was 29.7 wt% while that in [BMIM]Cl-NMTU (2:1) was only 17.0 wt%, at 343.15 K. The solubility of xylan in [AMIM]Cl-EG (2:1) could reach 40 wt% at 343.15 K. Even at 323.15 K, the solubility of xylan could reach 23 wt%, which was 15 wt% higher than that in the TU-based and NMTU-based DESs. Additionally, the solubilities of xylan in NMTU-based DESs ([EMIM]Cl-NMTU (2:1), [BMIM]Cl-NMTU (2:1), [HMIM]Cl-NMTU (2:1), [AMIM]Cl-NMTU (2:1))

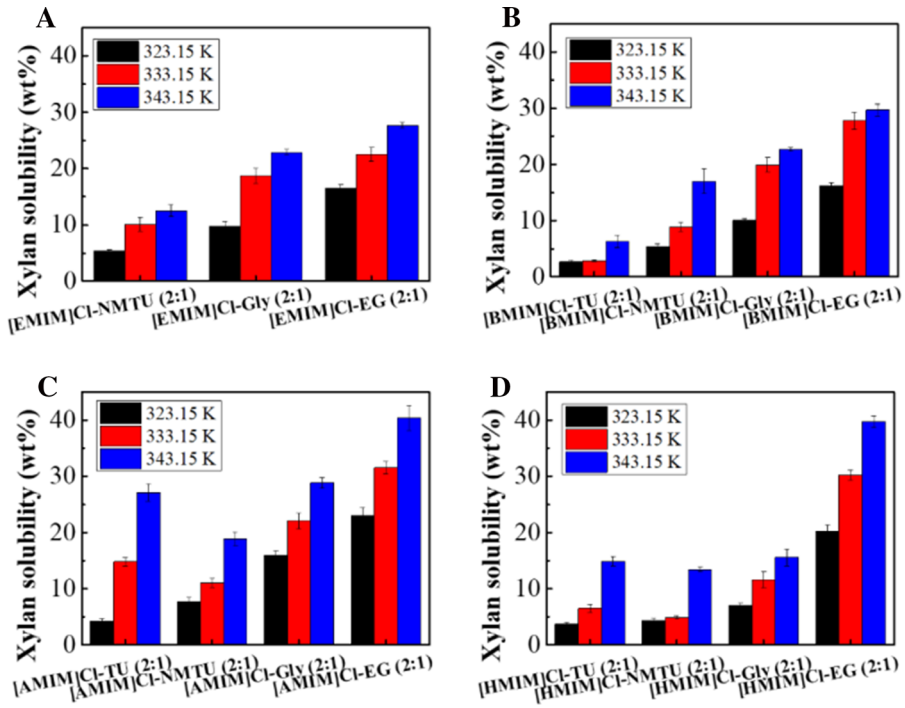


Fig. 4 The solubilities of xylan in the prepared DESs

are higher than that in the corresponding TU-based DESs.

From the chemical structure of the used HBDs (Fig. 1a), ethylene glycol and glycerol contained two and three hydroxyl groups, respectively, while thiourea and *N*-methylthiourea contain two amino groups, respectively. Therefore, it could be considered that when the HBD of the DESs contains hydroxyl groups, the solubilities of xylan in these DESs are higher than those in the DESs containing amino groups. The reason may be that the higher electronegativity of O than that of N resulted in the stronger hydrogen-bond formation ability of the HBDs (EG and Gly) with hydroxyl group than that of the HBDs (TU and NMTU) with amino group, which leaded to the high solubility of xylan in the DESs with HBD containing hydroxyl group. Meanwhile, the viscosities of the Gly-based and EG-based DESs were much lower than those of TU-based and NMTU-based DESs, which also contributed to the dissolution ability of the DESs with different HBDs. In addition, xylan had higher solubility in EG-based DESs than in Gly-based DESs, resulting from the higher viscosity of Gly-based DESs and the weaker hydrogen-bond formation ability between the Gly-based DESs and the xylan, which was caused by that glycerol was more likely to form intermolecular hydrogen bonds than ethylene glycol.

Correlation of xylan solubility with Kamlet-Taft parameters

To explore the dissolution mechanism, the Kamlet-Taft parameters of the prepared DESs were further measured (Table S11). Kamlet-Taft parameters were consisted of α, β and π* (Ab Rani et al. 2011; Jessop et al. 2012; Ke et al. 2018; Sun et al. 2014), which represent the hydrogen bond acidity (hydrogen-bond donating ability), the hydrogen bond basicity (hydrogen-bond accepting ability) and the polarity of the solvent, respectively. A statistical method whereby linear solvation energy relationship (LSER) was fitted by multiple linear regression to analyse the relationship between solubility and Kamlet-Taft parameters according to a reported method (Ab Rani et al. 2011; Gilbert et al. 2019), and the analysis method was given in "Material and methods" section. On the basis of the results shown in Table 2 and Figs. 5, S6, and S7, the solubility of xylan was not obviously correlated to the hydrogen-bond donating ability (α) and polarity (π*)

Table 2 Results of Kamlet-Taft LSER Fits for the dissolution of xylan in different kinds of DESs

DESs	Temperature (K)	Parameters	LSER equation	*p* values
IL-TU (2:1)	343.15	β	ln S = − 1.86 + 3.08β	XYZ_0 = 0.02, B = 0.01
IL-TU (3:1)	323.15	β	ln S = 0.48 + 0.65β	XYZ_0 = 0.03, B = 0.02
IL-NMTU (2:1)	333.15	β	ln S = − 0.46 + 2.32β	XYZ_0 = 0.05, B = 0.04
IL-Gly (1:1)	343.15	β	ln S = 2.07 + 0.93β	XYZ_0 = 0.05, B = 0.04
IL-Gly (1:1)	323.15	β	ln S = − 2.07 + 3.39β	XYZ_0 = 0.03, B = 0.03
IL-EG (2:1)	333.15	β	ln S = − 1.54 + 3.89β	XYZ_0 = 0.05, B = 0.04

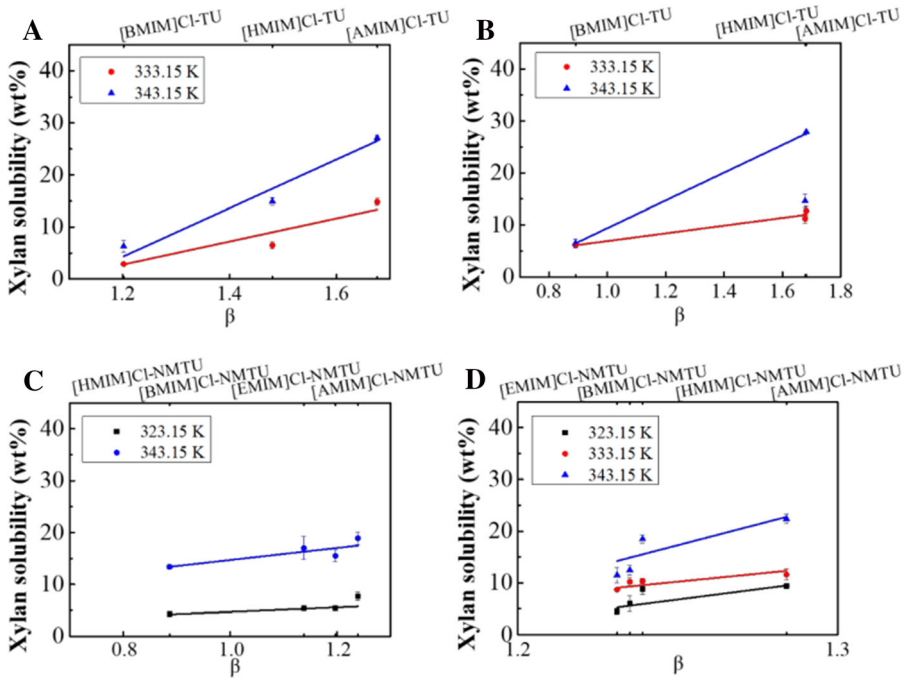


Fig. 5 The correlation of the xylan solubility and the value of β. **a** The β values of IL-TU (2:1). **b** The β values of IL-TU (3:1). **c** The β values of IL-NMTU (2:1). **d** The β values of IL-NMTU (3:1)

of DESs, but was positively correlated with the hydrogen-bond accepting ability of DESs, which was consistent with the results obtained in cellulose dissolution.

As shown in Fig. 5, the β values of [HMIM]Cl-NMTU (2:1), [BMIM]Cl-NMTU (2:1), [EMIM]Cl-NMTU (2:1) and [AMIM]Cl-NMTU (2:1) were 0.887, 1.138, 1.197 and 1.503, respectively. Meanwhile, the solubilities of xylan in these DESs were 4.3, 5.4, 5.4 and 7.7 wt%, respectively, at 323.15 K. These results indicated that the solubility of xylan in IL-NMTU (2:1) based DESs was correlated to their hydrogen-bond accepting ability. More importantly, increasing the molar amount of HBAs (ILs) would increase the β value of the corresponding DES, which was beneficial to the dissolution of xylan (Fig. 3). For example, the β values of [HMIM]Cl-NMTU (2:1) and [HMIM]Cl-NMTU (3:1) were 0.887 and 1.239, respectively. Correspondingly, the solubilities of xylan in [HMIM]Cl-NMTU (2:1) and [HMIM]Cl-NMTU (3:1) were 4.9 wt% and 10.4 wt% at 333.15 K, respectively.

Furthermore, the relationship between the xylan solubility and the β values of DESs formed by the

combination of ILs and polyol was analyzed. As shown in Figure S6, the solubility of xylan in [EMIM]Cl-EG (1:1) and [BMIM]Cl-EG (1:1) were similar although [BMIM]Cl-EG (1:1) had a higher β value than that of [EMIM]Cl-EG (1:1). The reason may be ascribed to that the viscosity of [BMIM]Cl-EG (1:1) was higher than [EMIM]Cl-EG (1:1), which would decrease the solubility of xylan. Meanwhile, Figs. 5 and S7 showed the solubility of xylan in [AMIM]Cl-based DESs at 333.15 K was not much higher than that at 323.15 K, but the xylan solubility was greatly improved when the temperature was raised to 343.15 K. It could be seen from Figure S8 that when the temperature was raised from 333.15 to 343.15 K, the percentage of viscosity reduction was lower than that as the temperature changed from 323.15 to 333.15 K, while the percentage of solubility increased at this case was significantly increased, indicating the relationship between viscosity, solubility and temperature. That was, the higher the temperature was, the lower the viscosity was, and thus, the higher the solubility was.

Reasons for the higher solubility in DESs than the corresponding IL

As discussed above, the solubility of xylan was higher in the synthesized DESs than the corresponding IL, which may be caused by two reasons. First, after forming DESs, the viscosities decreased significantly in comparison with the corresponding pure IL (Tables S1–S5). Lower viscosity was beneficial for xylan dissolution. Second, the β values of DESs were higher than that of the corresponding pure ILs, indicating that the combination of ILs and HBDs to form DESs could enlarge the hydrogen bond accepting ability, which was helpful to enhance the hydrogen-bond interaction between the solvent and the xylan, and thus the solubility of xylan in the DESs was increased.

Characterization of the regenerated xylan

The dissolved xylan could be easily regenerated from the DESs (employing [EMIM]Cl-NMTU (3:1) as the example) using ethanol as the anti-solvent. Several techniques, including FT-IR spectra, ^{1}H NMR and ^{13}C NMR, were conducted to confirm the integrity and purity of the regenerated xylan.

From the FT-IR spectra (Fig. 6), it was observed that the regenerated xylan had similar characteristic bands with the original xylan. For example, the band at 1043 cm^{-1} was the characteristic absorption of xylan, corresponding to the C–O–C stretching vibration of pyranoid-ring xylans (Morais et al. 2018). Meanwhile, the strong band at 896 cm^{-1} corresponded to the glyosidic bond β-(1 → 4) between the xylopiranose units of the main xylan chain (Morais et al. 2018),[22] while the bands at 1461 and 1375 cm^{-1} belonged to the stretching peak of $–CH_2–$ and C–O, respectively (Sun et al. 1998). Moreover, the strong absorption peaks at 3384 and 2918 cm^{-1} were O–H stretching vibration and C–H stretching, respectively. Additionally, the width of the O–H stretching vibration peak (3384 cm^{-1}) of the regenerated xylan narrowed, indicating that the intermolecular and intramolecular hydrogen bonding interactions between xylan were reduced in the process of dissolution and regeneration (Kasprzak et al. 2019; Li et al. 2019).

Furthermore, ^{1}H NMR and ^{13}C NMR were employed to characterize the original and regenerated xylan. The two simples showed similar ^{1}H NMR spectra (Fig. 7a). It should be pointed out that the peaks at 3.23 and 5.32 ppm were observed for both samples (Fig. 7b, c), which were the typical resonances of the 4-OCH_3 and H-1 in the 4-*O*-methyl-D-glucuronic aid moiety (4-*O*-methy-α-D-GlcpA) (Evtuguin et al. 2003; Morais et al. 2018), suggesting the dissolution process did not affect the substituted glucuronic aid moiety. Meanwhile, in the ^{13}C NMR spectra for both samples (Figure S9), all the typical

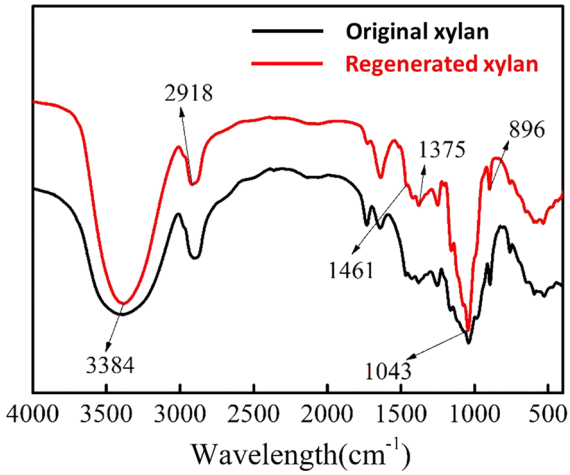


Fig. 6 FT-IR spectra of the original and regenerated xylan

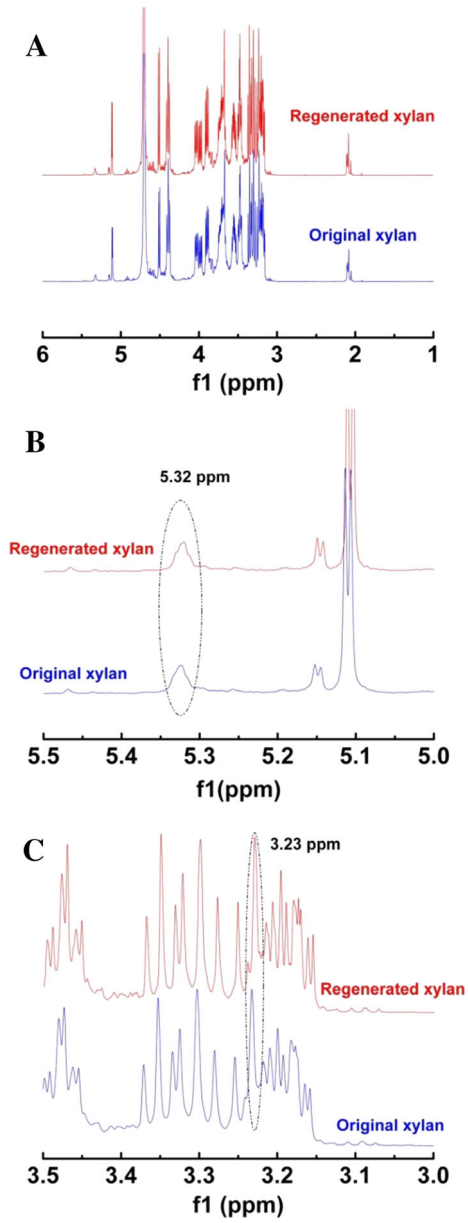


Fig. 7 ^{1}H NMR spectra for the original and regenerated xylan (**a**) and the corresponding enlarged areas (**b**, **c**)

carbons were observed in (1 → 4)-β-D-Xylp and 4-*O*-methy-α-D-GlcpA units (Morais et al. 2018; Sun et al. 2012), which were consistent with the results in ^{1}H NMR spectra.

Additionally, the acetylation degrees in the original and regenerated xylan were 0.14 and 0.12, respectively, calculated from the ^{1}H NMR spectra (Lee et al. 2014).Meanwhile, a slight decrease of uronic acids content (from 10 to 8 wt%) was observed for the regenerated xylan compared with the original xylan (Bastos et al. 2015; Morais et al. 2018).

From the above characterizations, we could conclude that the structure of the regenerated xylan almost unchanged during the process of dissolution and regeneration although 4-*O*-methy-α-D-GlcpA units were partially removed.

Conclusions

In summary, several IL-based DESs were designed and synthesized using four ionic liquids as the hydrogen bond acceptors and thiourea, N-methylthiourea, glycerol and ethylene glycol as the hydrogen bond donors. It was found that these synthesized IL-based DESs showed good solubility of xylan, especially for [AMIM]Cl-EG (2:1) with the maximum solubility of 40.4 wt% at 343.15 K compared with those solvents of previous literature. Meanwhile, the xylan solubility was mainly positively correlated with the hydrogen bond accepting ability of DESs. More importantly, the solubility of xylan in these IL-based DESs was much higher than that in the corresponding pure IL owing to the significant decrease of viscosity and the enhanced hydrogen bond accepting ability after the formation of DESs. We believe that the developed DESs have great potential in the separation, dissolution and even conversion of lignocellulose in the future, and formation of DESs is a promising strategy to improve the ability of IL to dissolve biomass, including cellulose, hemicellulose, and lignin.

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